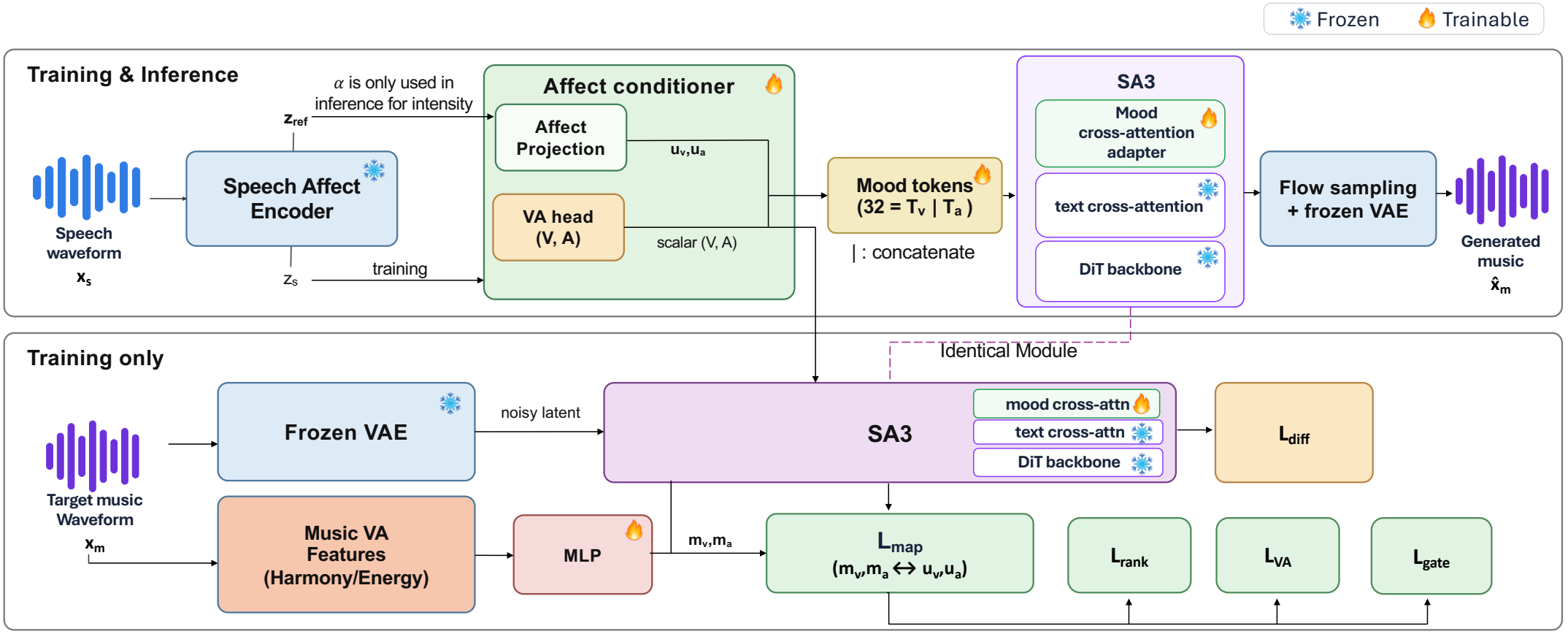

Frozen
Trainable
Training & Inference
Speech waveform
x_s
Speech Affect Encoder
z_{ref}
z_s
α is only used in inference for intensity
training
Affect conditioner
Affect Projection
u_v, u_a
VA head (V, A)
scalar (V, A)
Mood tokens (32 = T_v | T_a)
| : concatenate
SA3
Mood cross-attention adapter
text cross-attention
DiT backbone
Flow sampling + frozen VAE
Generated music
$\hat{x}_m$
Training only
Identical Module
Target music Waveform
x_m
Frozen VAE
noisy latent
SA3
mood cross-attn
text cross-attn
DiT backbone
L_{diff}
Music VA Features (Harmony/Energy)
MLP
m_v, m_a
L_{map} ($m_v, m_a \leftrightarrow u_v, u_a$)
L_{rank}
L_{VA}
L_{gate}